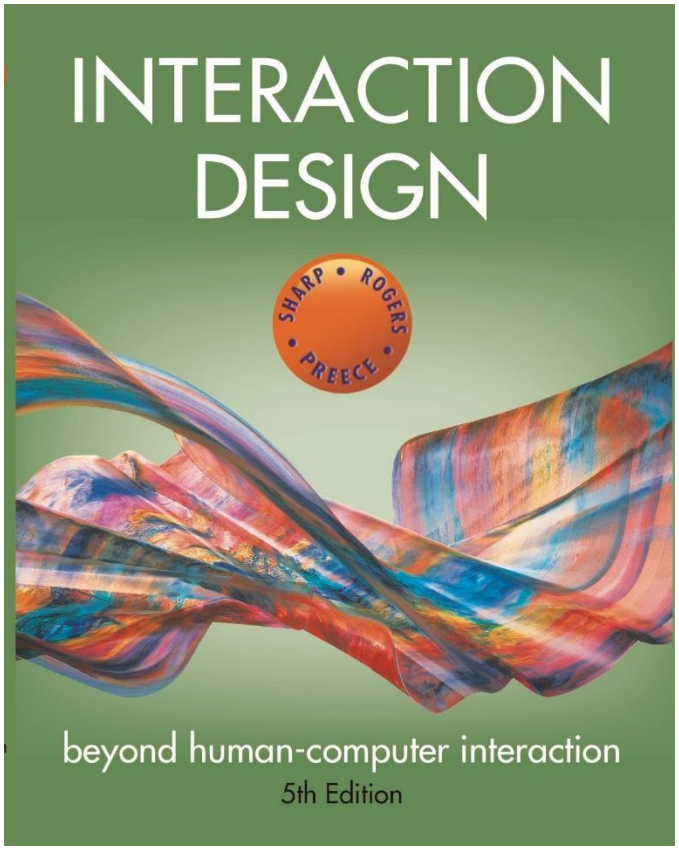
Week 11
Lecture 14
Data at Scale

Dr. Usama Mir
umir@wlu.ca



Chapter 10
DATA AT SCALE

What is data at scale?

- Data at scale is also known as big data
- It describes numbers, images of people, places and things, conversations, video recordings, sensor data, and so on
- Huge potential for solving problems
- But you must know how to collect, analyze, and communicate findings
- Also involves dangers to people's privacy

Collecting data at scale?

- Easy to collect masses of data, but then **what do you do with it**?

- What are the **rules and policies** that apply when collecting data about people? For example, tracking people's activities as they move through Heathrow and other airports for security reasons may also violate their privacy.

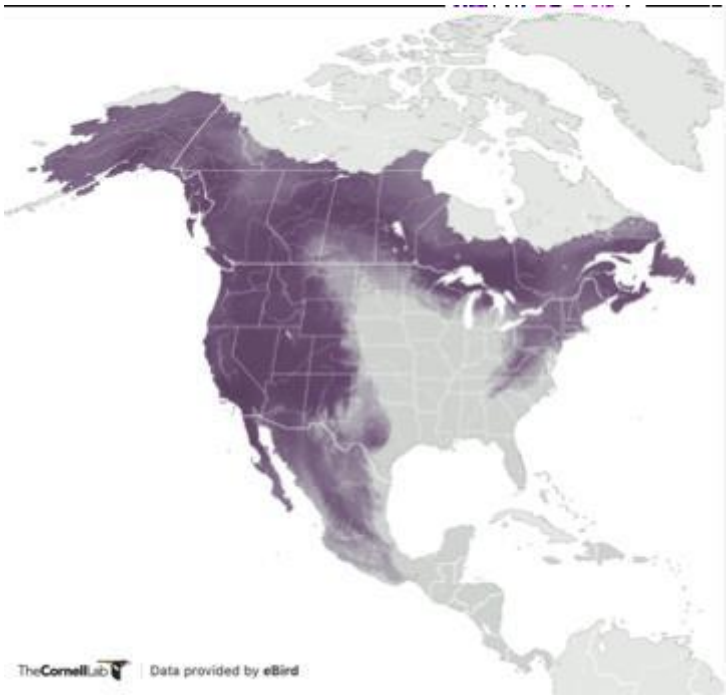
- Also, scraping and crowdsourcing data, personal data collection directly and indirectly through devices, and sentiment and social network analysis.

Collecting personal data at Heathrow airport

Source: Mark Zakian / Alamy Stock Photo

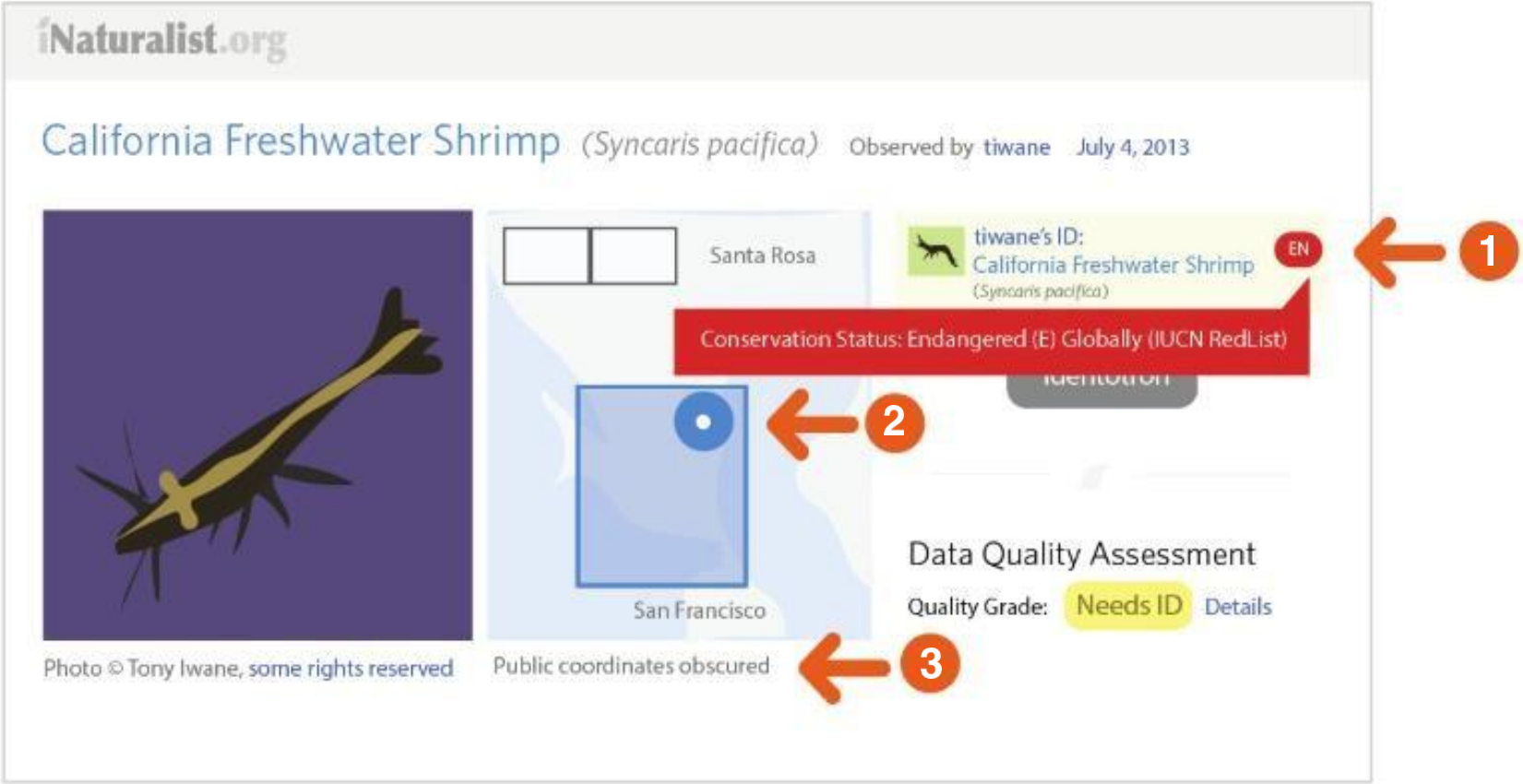
Approaches for Data Collection: Crowdsourcing data

- Clicking on this link (Common Raven) will allow you to see how the abundance of raven changes during each week of the year. (Purple indicates high abundance and yellow indicates low abundance.)



Source: https://ebird.org/science/status-and-trends/comrav

Approaches for Data Collection: Citizen science and privacy



Source: iNaturalist.org

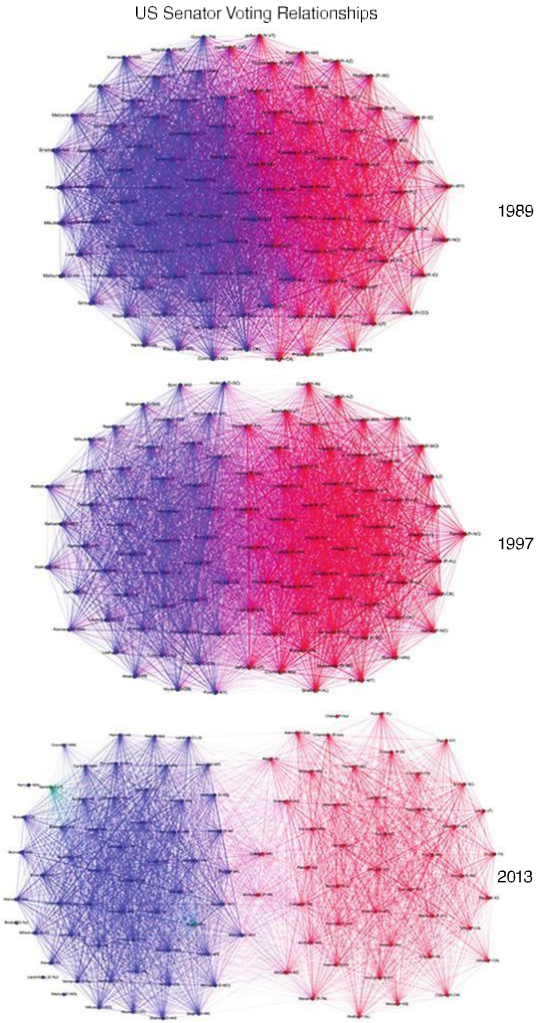
Approaches for Data Collection: Sentiment Analysis

- Sentiment analysis is a technique that is used to infer the effect of what a group of people or a crowd is feeling or saying
- Some sentiment systems provide more qualitative measures by identifying if the positive or negative sentiment is associated with a specific feeling, for example anger, sadness, or fear (negative feelings) or happiness, joy, or enthusiasm (positive feelings)
- Info is extracted from people's tweets, posts, reviews, and facial expressions
- Algos are then applied
- Tools such as DisplayR and CrowdFlower can be used

(c)

Approaches for Data Collection: Social network analysis

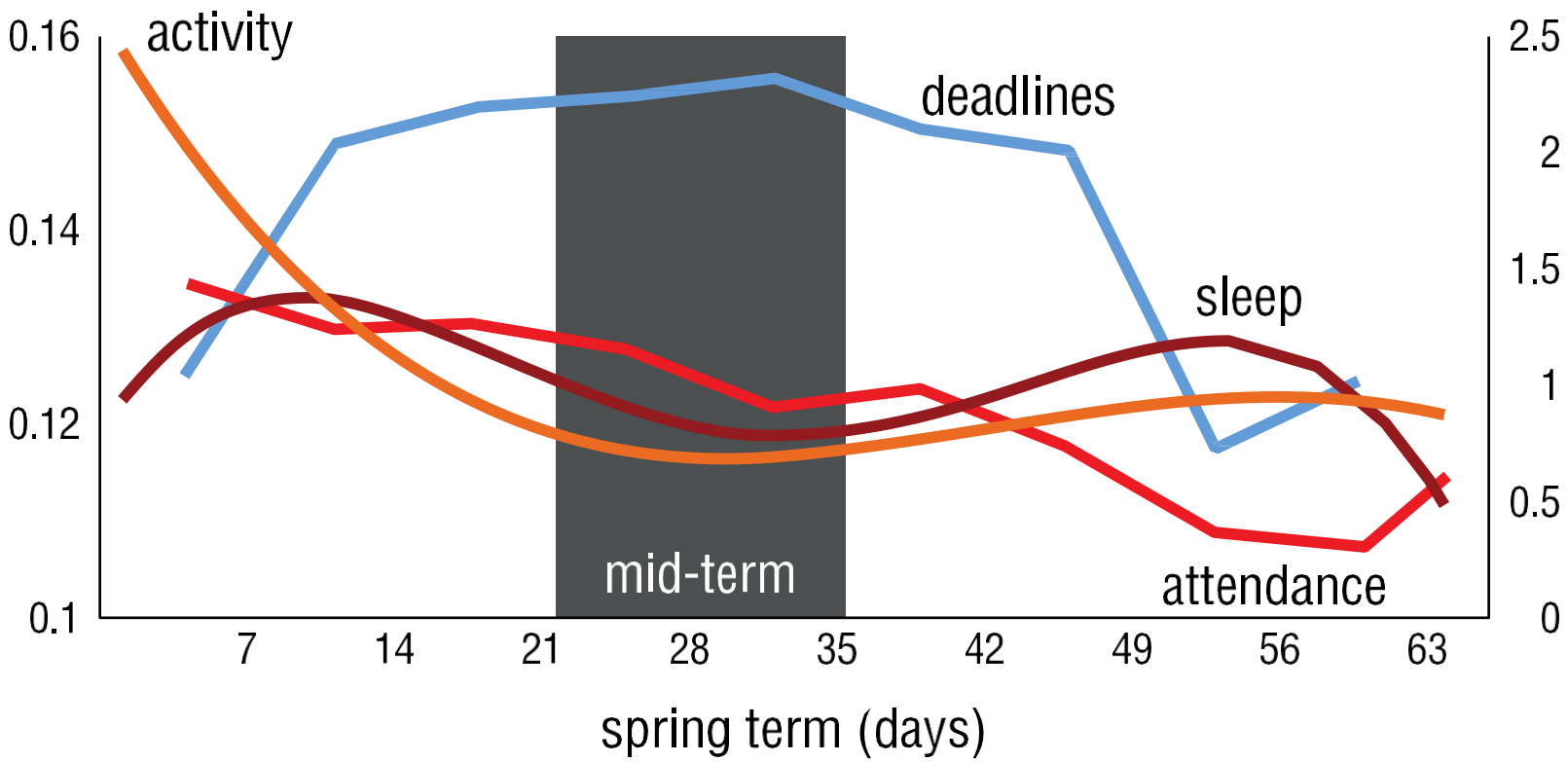
- US Senators voting relationships in 1989 (top), 1997 (middle), and 2013 (bottom)
- Red represents Republicans and blue represents Democrats



Source: Forbes

Approaches for Data Collection: Students' activity (1)

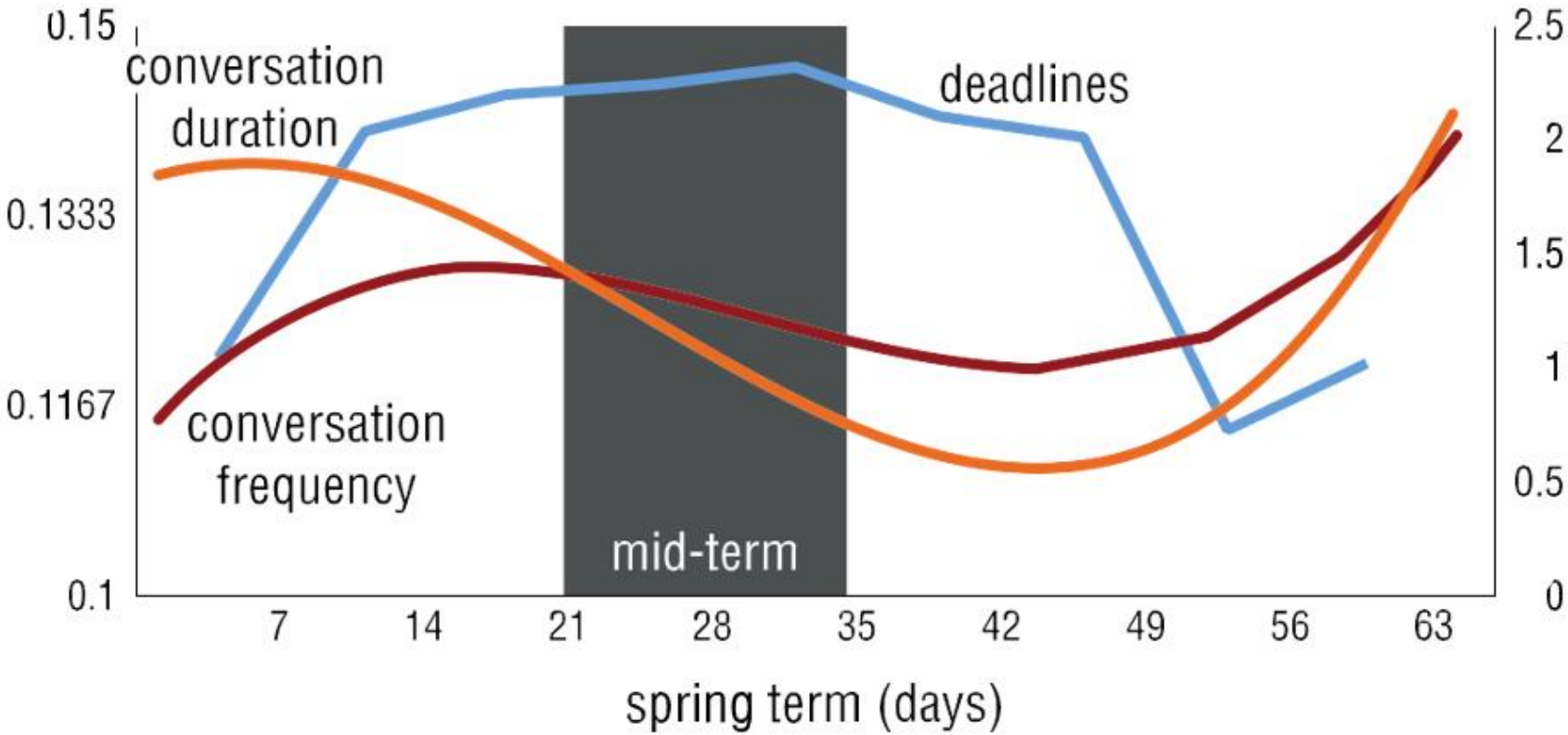
Combining multiple sources of data



Source: StudentLife Study

Approaches for Data Collection: Students' activity (2)

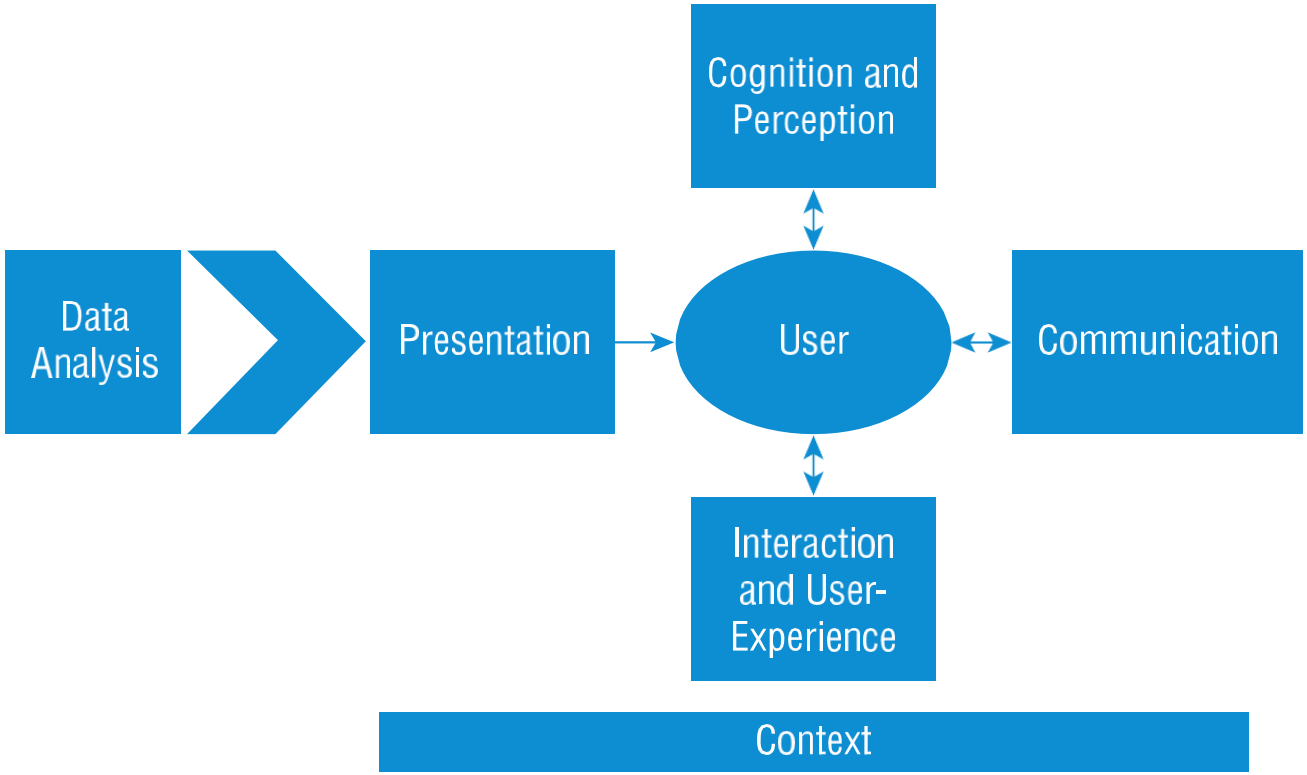
Combining multiple sources of data



Source: StudentLife Study

Visualizing and exploring data

- A simplified path for data to be meaningful

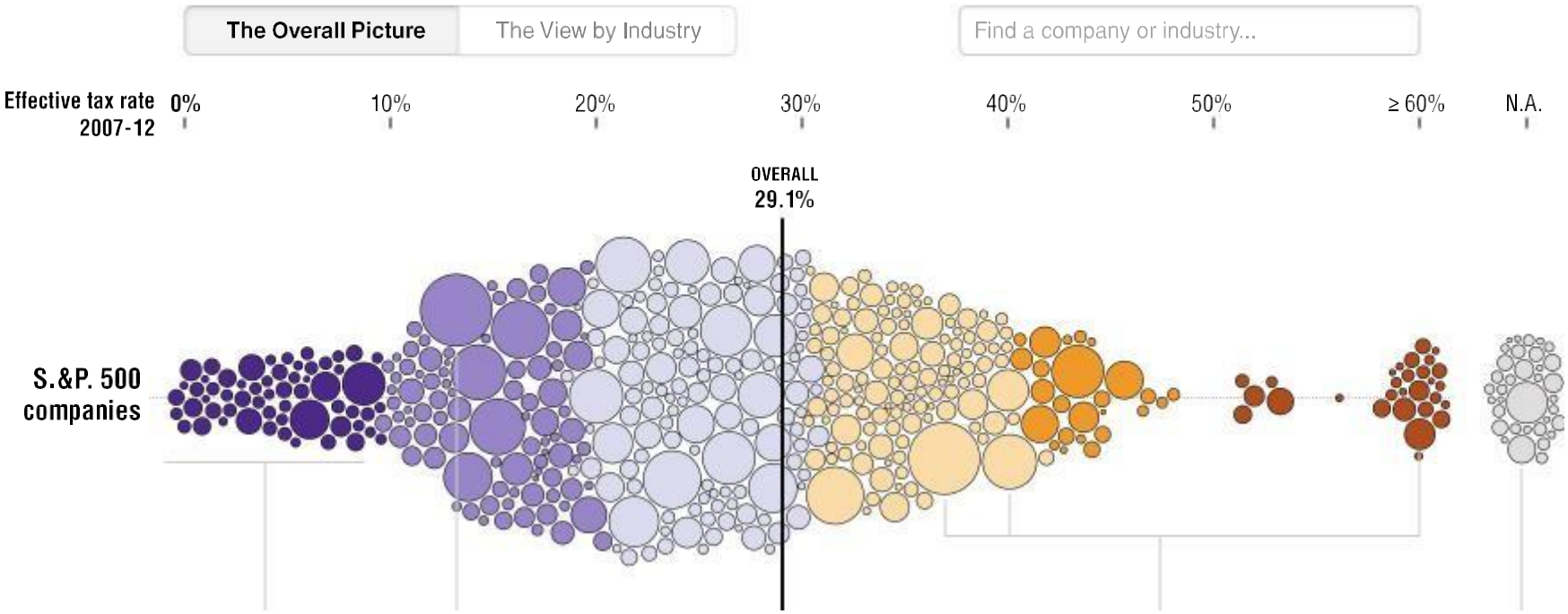


Source: Used courtesy of Dr. Artur Lugmyr

Visualizing and exploring data: Market Maps

Source: Used courtesy of FINVIZ

Visualizing and exploring data: D3 Approach

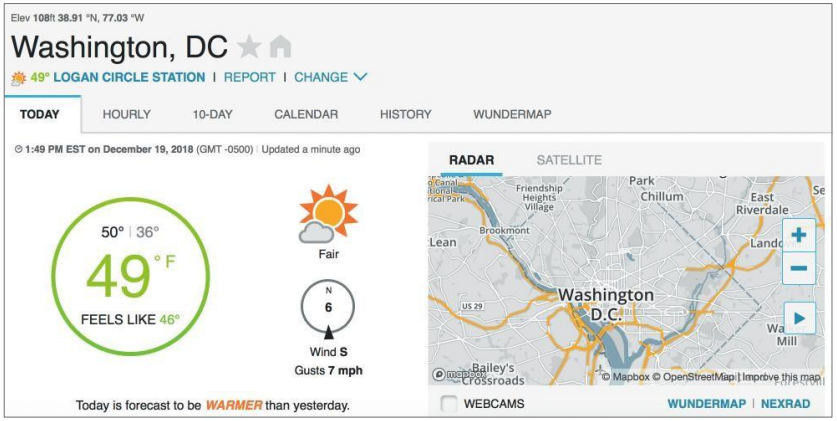


[Across U.S. Companies, Tax Rates Vary Greatly - Interactive Feature - NYTimes.com]

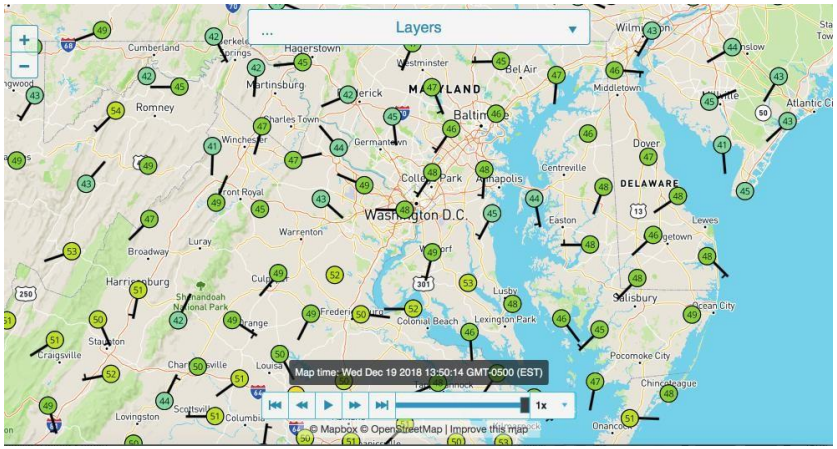
D3: Data-Driven Documents

Visualizing and exploring data: Weather data

- (a) Actual weather data
- (b) Wundermap of data from the same area at the same time



(a)

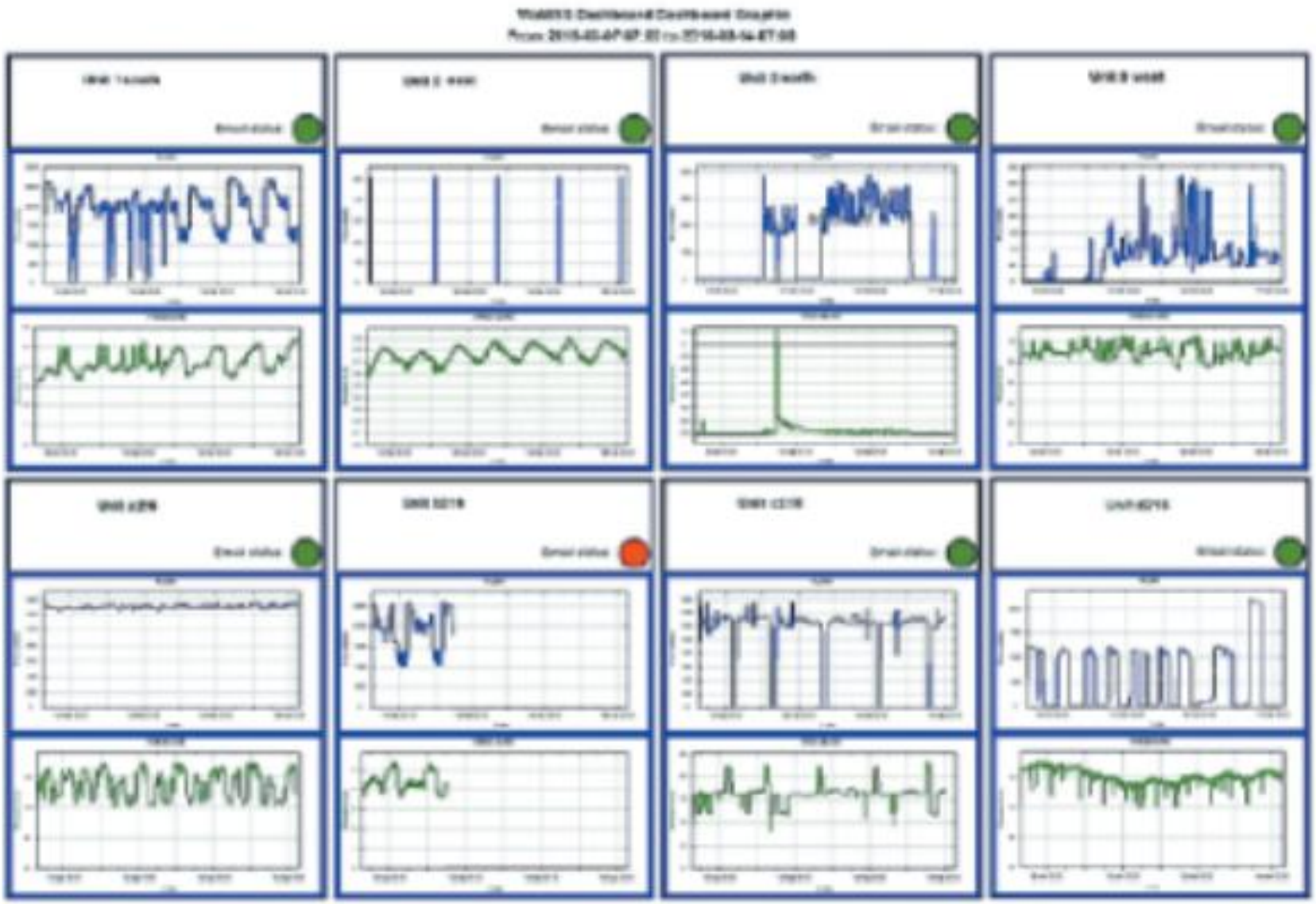


(b)

Source: Weather Underground

Visualizing and exploring data: Different dashboards (1)

Visualizing and exploring data: Different dashboards (2)



Visualizing and exploring data: DeepCam's face-tracking software

Source: DeepCam

Ethics: Data ethics principles (FATE)

Fairness:
Is the treatment just and without favoritism or discrimination?

Accountability:
Is the data accurate and correct?

Transparency:
Are the decisions being made by a system visible?

Explainability:
Can people understand the explanations provided by the system?

The goal of **FATE** is to make decisions made by computer systems just and fair.

And then there is Privacy as well

Ethics: IEEE/ACM Code of Ethics

Code of Ethics

IEEE Governing Documents | IEEE

Further Reading on Big Data Analysis

https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8485391

Summary

- This chapter discussed how data at scale is collected, analyzed, and communicated.
- The data can be qualitative or qualitative.
- It comes from different sources, for example, sensors, social media, documents, and facial recognition data.
- Privacy is a big concern because data from different sources can be brought together to reveal patterns of behavior.
- The FATE frameworks suggests ethical principles to guide UX design: Fairness, Accountability, Transparency, and Explainability.